Alicia de Cubas Yarza

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ABOUT ME

I am a final-year Computer Science for Games student at Sheffield Hallam University, with a background in videogame programming. I enjoy creating responsive gameplay systems and intuitive player interfaces, as well as exploring the engine and graphics technologies behind games. Through projects using Unreal Engine, Unity, C++, OpenGL and Vulkan, I have developed experience in gameplay programming, rendering and technical problem-solving. I enjoy collaborating with others and turning ideas into playable experiences.

TECHNICAL SKILLS

- Strong programming skills in C++, C#, C, Python, Lua, GLSL, HLSL and Kotlin
- Experience with game development in Unreal Engine and Unity, including gameplay systems, UI implementation, audio systems and interactive mechanics
- Experience with graphics APIs and rendering technologies, including OpenGL, Vulkan, shaders, lighting, materials, geometry and particle systems
- Understanding of core gameplay programming concepts, including player controllers, enemy behaviour, combat systems, inventory systems, HUD design and interaction logic
- Experience with procedural generation and environment creation, including terrain generation and scalable world-building systems
- Understanding of software engineering principles, including version control, debugging, profiling, problem solving and iterative development with Git and GitHub
- Experience working with Windows and Linux operating systems
- Strong understanding of user-centred design, focusing on intuitive gameplay, readable interfaces and engaging player experiences
- Able to work effectively in team-based projects, collaborating on technical implementation, communication and project delivery

EDUCATION

- **Computer Science for Games** (2026-2027)
  BSc (Honours)
  Sheffield Hallam University

- **HND in Computing (Game Programming)** (2023-2026)
  HND Degree
  ESAT

PROJECTS

- **NAOS01: The Quiet Sector | Unreal Engine** (2026)

- Developed as a final university project: a stealth game focused on tension, exploration and responsive gameplay systems.
- Implemented the HUD, player interaction systems, radar/sonar and audio integration.
- Developed NPC behaviour to support the game's stealth gameplay.

▪ **Graphics Engine | C++ / OpenGL** (2025)
- Contributed to a team-built 3D graphics engine at ESAT.
- Worked on procedural terrain, GPU buffers, materials, textures, particles, geometry and lighting.
- Contributed to the engine's rendering systems and visual features.

▪ **Procedural Terrain | Unity** (2025/2026)
- Developed a Unity project focused on procedural terrain and real-time environment generation.
- Built terrain-generation and world-building functionality for creating dynamic environments.

▪ **Bullet Hell Game | Unity** (2025)
- Developed a bullet hell game as part of a team project.
- Implemented the player controller, enemy firing patterns, HUD and parallax background movement.
- Contributed to the core combat and gameplay experience.

▪ **Weapon & Inventory Demo | Unreal Engine** (2025)
- Built the player weapon system and inventory flow for a team-based shooter prototype.
- Implemented weapon management and progression logic.
- Developed gameplay systems supporting the player's combat experience.

▪ **OpenGL/ Vulkan Graphics Engine | C++ / OpenGL / Vulkan** (2025)
- Developed a graphics engine in a team of three, supporting both OpenGL and Vulkan.
- Contributed foundational OpenGL components and implemented the Vulkan backend.
- Built a compatibility layer connecting the two graphics APIs.

▪ **Voice-Controlled Stealth Game | Unity** (2025)
- Created a two-person Unity game exploring microphone-based movement as an alternative to keyboard input.
- Implemented voice-controlled movement, with speaking volume affecting the player's speed.
- Designed the stealth objective around reaching the kitchen before time runs out without alerting the player's parents.

LANGUAGES

- Spanish (Native)
- English (C1 - IELTS Band Score 7.0 / 8.0)
- Valenciano (C1)